WENHUI CUI

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Education

Northwestern Polytechnical University — **Xi'an, China**
Master, Software Engineering — Sep 2022 – Apr 2025

Southwest Jiaotong University — **Chengdu, China**
Bachelor, Network Engineering — Sep 2018 – Jun 2022

- National Encouragement Scholarship (2019, 2021, for top 10%)

Experience

General Software Engineer — **Jun 2025 – Present**
Huawei Technologies — *Shanghai, China*

- Contributed to the stabilization of parallel compilation in the Rust compiler, collaborating with the Rust community to resolve concurrency and determinism issues
- Designed and implemented benchmarking suites for the Rust standard library and Daft on ARM platforms, enabling systematic performance analysis and regression tracking
- Optimized core components through assembly-level analysis, LLVM tuning, vectorization, and reduced dynamic dispatch overhead
- Developed AI Agent–driven workflows for automated C/C++ to Rust migration, improving developer productivity and migration reliability

Software Engineer Intern — **Jun 2023 – Sep 2023**
Kmerit — *Suzhou, China*

- Refactored a Spring Boot-based financial analytics system using Apache Flink, reducing computation time for 100,000 assets from 60s to 20s through parallel stream processing
- Implemented hierarchical aggregation pipelines with Apache Flink and streamed real-time metrics to frontend services via Apache Kafka

Projects

Rust Benchmarking Platform | Rust, postgresql, Vue

- Built a Rust benchmarking platform based on rustc-perf for the Rust standard library and Daft, including automated benchmarking pipelines and a Vue-based visualization frontend
- Performed low-level performance analysis using perf, flamegraphs, and ARM assembly inspection to identify optimization opportunities across 5+ Daft interfaces
- Applied LLVM tuning, manual vectorization, and dispatch optimizations, improving Daft performance on ARM by up to 8%

Rust Parallel Compilation Stabilization | Rust, GitHub CI

- Contributed multiple upstream PRs to the Rust compiler, fixing deadlocks, data races, and non-deterministic compilation behavior in parallel compilation
- Expanded compiler test coverage with 20+ parallel compilation test cases and CI support improvements
- Collaborated with Rust compiler maintainers on design discussions and implementation details to advance parallel compilation stabilization

Web Novel Reader | Rust, Flutter, Lua

- Developed a cross-platform web novel reader using Flutter and Rust, with an Actor-model backend for high-concurrency task scheduling
- Built a sandboxed Lua runtime using mlua, supporting extensible parsing scripts, authentication flows, pagination, and dynamic content extraction
- Implemented local caching and reading progress synchronization for resilient offline reading experiences

Skills

Languages: Rust, Java, Python
Systems & Performance: ARM performance optimization, assembly analysis, LLVM tuning, concurrent systems, JVM troubleshooting
Infrastructure: Apache Flink, Apache Kafka, MySQL, PostgreSQL
Tools: Linux, Git, GitHub Actions, perf, flamegraph